

Master Math for JEE Main & JEE Advanced

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JEE Mains 2020 Jan Chapter wise Question Bank

Wave Optics

Q1

In a single slit diffraction set up, second minima is observed at an angle of 60° . The expected position of first minima is

एकल छिद्र विवर्तन प्रक्रिया में, 2nd निम्निष्ठ (minima) 60° के कोण पर दिखाई देता है, तो 1st निम्निष्ठ (minima) की अपेक्षित स्थिति होगी –

- (1) 25° (2) 20° (3) 30° (4) 45°

7th Jan Morning

Sol

(1)

For 2nd minima द्वितीय निम्निष्ठ के लिए

$$d \sin \theta = 2\lambda$$

$$\sin \theta = \frac{\sqrt{3}}{2} \text{ (given)}$$

$$\Rightarrow \frac{\lambda}{d} = \frac{\sqrt{3}}{4} \dots (i)$$

So for 1st minima is

इसलिए प्रथम निम्निष्ठ के लिए

$$d \sin \theta = \lambda$$

$$\sin \theta = \frac{\lambda}{d} = \frac{\sqrt{3}}{4} \text{ (from equation (i) (समीकरण (i) से)}$$

$$\theta = 25.65^\circ \text{ (from sin table) (sin तालिका से)}$$

$$\theta \approx 25^\circ$$

Q2

If 10% of intensity is passed from analyser, then, the angle by which analyser should be rotated such that transmitted intensity becomes zero. (Assume no absorption by analyser and polarizer).

यदि एक ध्रुवक से 10% तीव्रता गुजरती हो तो ध्रुवक को कितने कोण से घुमाये जाये कि निर्गत तीव्रता शून्य हो जाये। (मान लीजिए की ध्रुवक से अवशोषण शून्य है)

- (1) 60° (2) 18.4° (3) 45° (4) 71.6°

7th Jan Morning

Sol

(B)

$$I = I_0 \cos^2 \theta$$

$$\frac{I_0}{10} = I_0 \cos^2 \theta$$

$$\cos \theta = \frac{1}{\sqrt{10}} = 0.31 < \frac{1}{\sqrt{2}} \text{ which is } 0.707$$

So $\theta > 45^\circ$ and तथा $90 - \theta < 45^\circ$ so only one option is correct इसलिए केवल एक विकल्प सही होगा।

i.e. अर्थात् 18.4°

angle rotated should be घुमाया हुआ कोण $= 90^\circ - 71.6^\circ = 18.4^\circ$

Q3

In YDSE, separation between slits is 0.15 mm, distance between slits and screen is 1.5 m and wavelength of light is 589 nm, then fringe width is

YDSE में छिद्रों के मध्य दूरी 0.15mm है, छिद्रों तथा पर्दे के मध्य दूरी 1.5m है तथा प्रकाश की तरंगदैर्घ्य 589 nm है, तो फ्रिंज चौड़ाई होगी।

(1) 5.9 mm

(2) 3.9 mm

(3) 1.9 mm

(4) 2.3 mm

7th Jan Evening

Sol

(1)

$$\beta = \frac{\lambda D}{d} = \frac{589 \times 10^{-9} \times 1.5}{0.15 \times 10^{-3}} = 5.9 \text{ mm}$$

Q4

In YDSE path difference at a point on screen is $\frac{\lambda}{8}$. Find ratio of intensity at this point with maximum intensity.

यंग के द्विक स्लिट प्रयोग में पर्दे पर स्थित किसी बिन्दु पर पहुँचने वाली तरंगों के बीच पथान्तर $\frac{\lambda}{8}$ है। इस बिन्दु पर प्रकाश की तीव्रता और अधिकतम तीव्रता का अनुपात होगा।

(1) 0.853

(2) 0.533

(3) 0.234

(4) 0.123

8th Jan Evening

Sol

(1)

$$I = I_0 \cos^2 \left(\frac{\Delta\phi}{2} \right)$$

$$\frac{I}{I_0} = \cos^2 \left[\frac{\frac{2\pi}{\lambda} \times \Delta x}{2} \right] = \cos^2 \left(\frac{\pi}{8} \right); \quad \frac{I}{I_0} = 0.853$$

Q5

Three waves of same intensity (I_0) having initial phases $0, \frac{\pi}{4}, -\frac{\pi}{4}$ rad respectively interfere at a point.

Find the resultant Intensity

समान तीव्रता (I_0) की तीन तरंगों की प्रारम्भिक कला क्रमशः $0, \frac{\pi}{4}, -\frac{\pi}{4}$ रेडियन है। तीनों एक बिन्दु पर व्यतिकरण उत्पन्न

करती है। इनकी परिणामी तीव्रता ज्ञात करो।

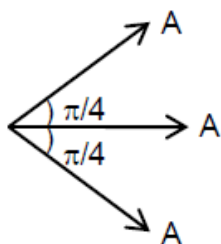
(1) I_0

(2) 0

(3) $5.8 I_0$ (4) $0.2 I_0$

Sol

(3)



$$A_{\text{res}} = (\sqrt{2} + 1)A$$

$$I_{\text{res}} = (\sqrt{2} + 1)^2 I_0$$

$$= (3 + 2\sqrt{2}) I_0 = 5.8 I_0$$

Q6

Wave Optics

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In YDSE pattern with light of wavelength $\lambda_1 = 500\text{nm}$, 15 fringes are obtained on a certain segment of screen. If number of fringes for light of wavelength λ_2 on same segment of screen is 10, then the value of λ_2 (in nm) is—

9th Jan Evening

Sol

750 nm

$$15 \times 500 \times \frac{D}{d} = 10 \times \lambda_2 \times \frac{D}{d}$$

$$\lambda_2 = 15 \times 50 \text{ nm}$$

$$\lambda_2 = 750 \text{ nm}$$



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